Struct hash

Structural hashing finds functionally identical subexpressions , the idea behind FRAIG and AIG strashing

The idea

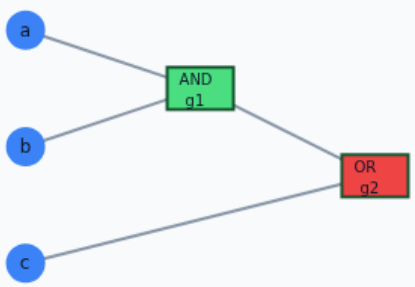
- Spec has 2 unique gates

Hash signatures

STRUCTURAL HASH

Hash signatures

Step 1 / 5 · idea · module02-03-structural-hash



Explanation

FRAIG-lite hashes each gate by (op, sorted child hashes).

- structural

```
not functional
```

Hash signatures

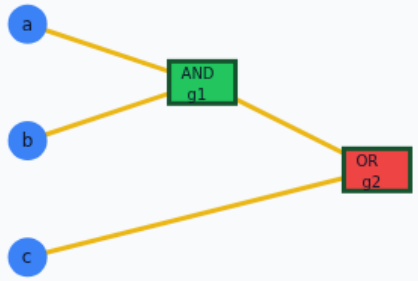
Spec hashes

STRUCTURAL HASH

Spec hashes

Step 2 / 5 · spec · module02-03-structural-hash

{"totalGates":2,"uniqueHashes":2,"redundant":0,"gateHashes"



Explanation

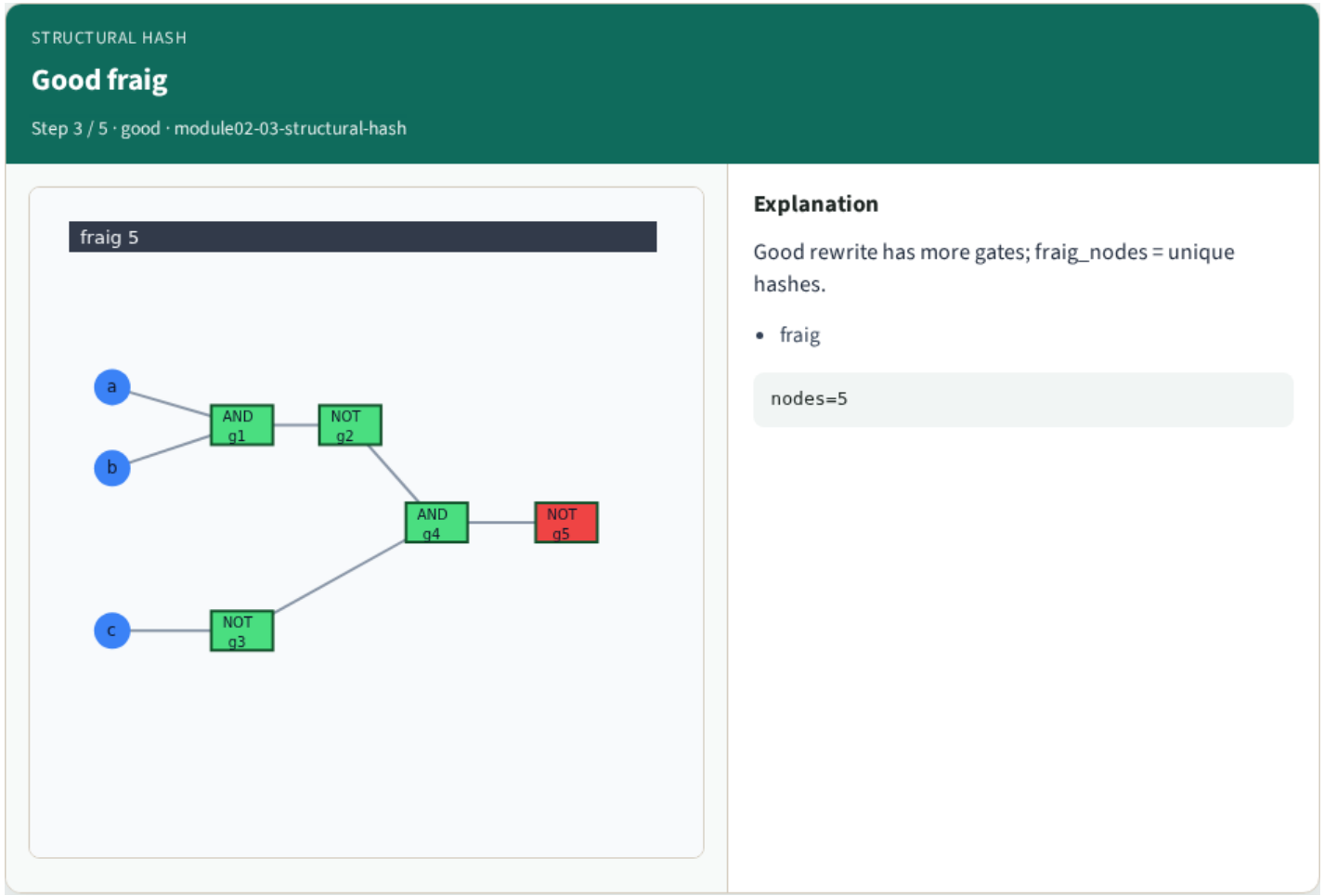
Spec has 2 gates → 2 unique hashes on the toy.

- unique

n=2

Spec hashes

Good fraig



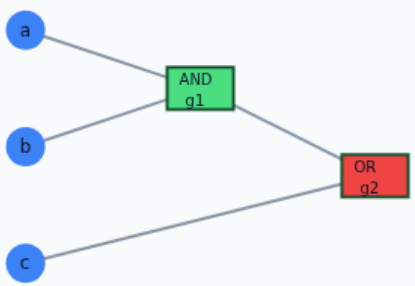
Good fraig

Not enough alone

STRUCTURAL HASH

Not enough alone

Step 4 / 5 · limit · module02-03-structural-hash



Explanation

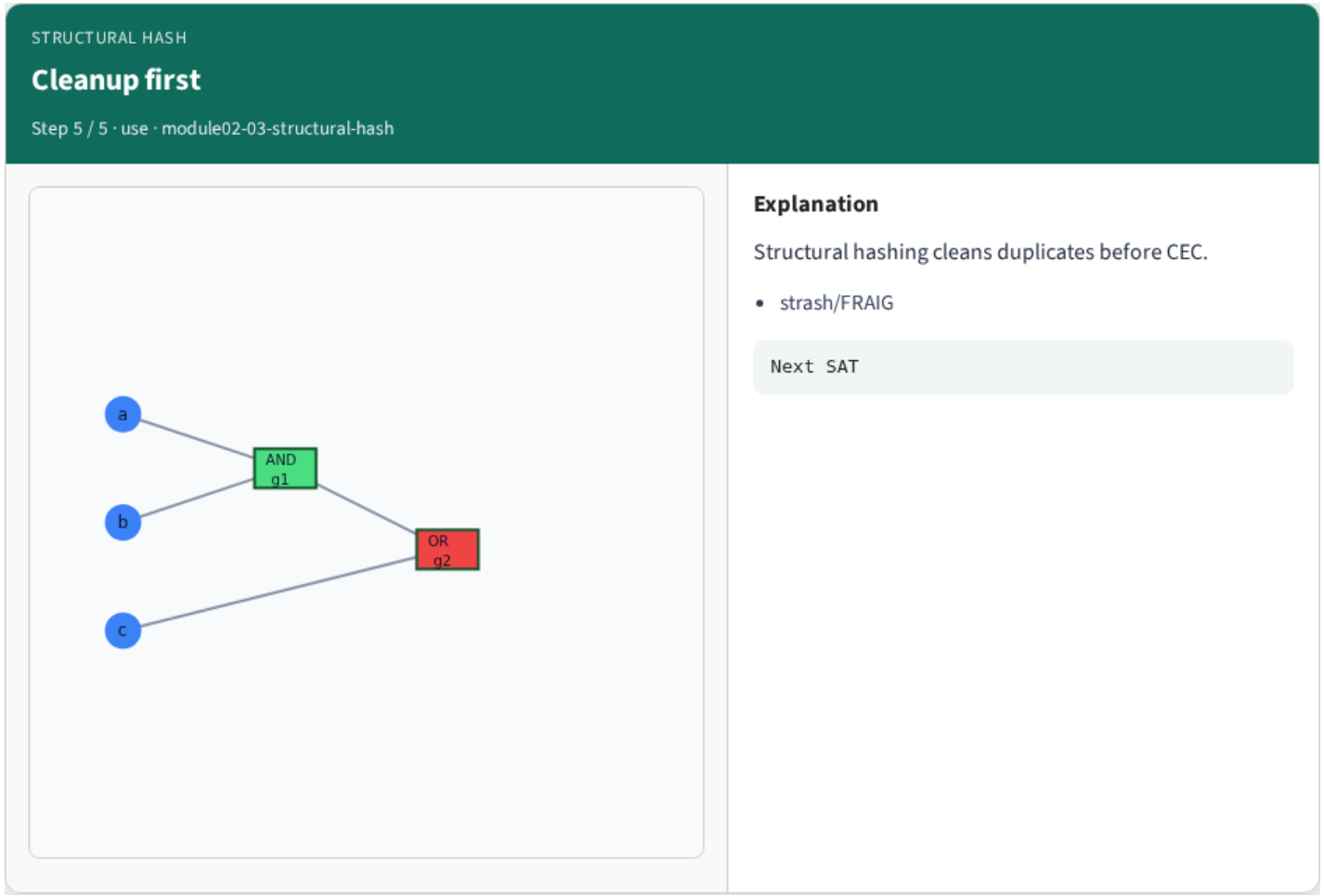
Different structures can still be functionally equal.

- need SAT/TT

```
literacy
```

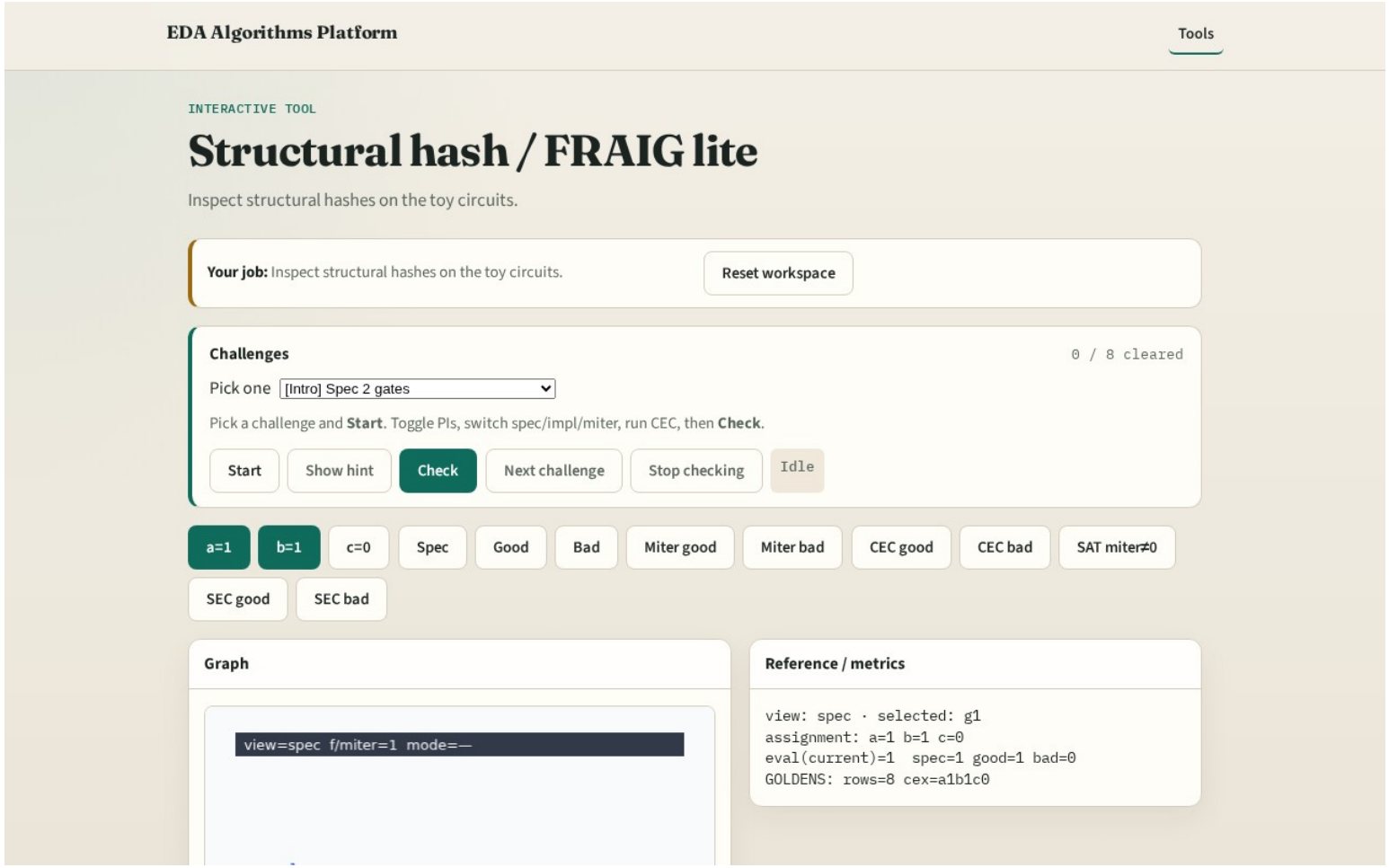
Not enough alone

Cleanup first



Cleanup first

Browser lab track



Browser lab starter

Implement track

- Call structural_hash_stats(spec) and fraig_lite(impl_good)

Pitfalls

- Hashing by gate id instead of structure
- Expecting full industrial FRAIG fixpoint

Your turn

- Next: SAT / CNF encoding lite